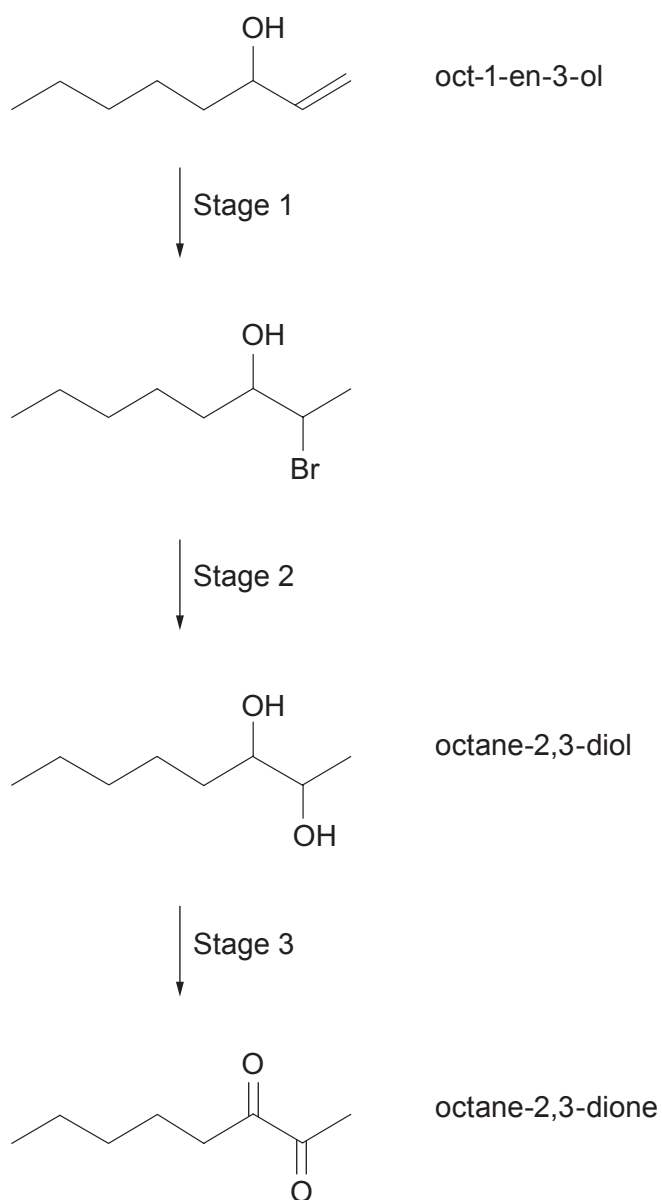


SECTION B

Answer all questions in the spaces provided.

6. (a) Oct-1-en-3-ol is a compound that attracts biting insects such as mosquitoes. It is contained in human breath and sweat and is also produced by some mushrooms.

A sequence of reactions starting with oct-1-en-3-ol is shown below.



- (i) Give the molecular formula of oct-1-en-3-ol. [1]

.....

- (ii) State a reagent that can be used for Stage 2 and name the type of reaction mechanism taking place. [2]

Reagent(s)

Type of reaction mechanism

- (iii) State a reagent that can be used for Stage 3. [1]

.....

- (b) An isomer of oct-1-en-3-ol was reacted with alkaline iodine and produced a yellow precipitate.

- (i) Give the formula of the yellow precipitate. [1]

.....

- (ii) Suggest a structural formula for this isomer and identify the group that enables it to produce this yellow precipitate. [2]

Structural formula

Group

- (c) Octane-2,3-dione is a pale yellow liquid and octane-2,3-diol is a colourless material.

- (i) State a reagent that can be used to produce octane-2,3-diol from octane-2,3-dione. [1]

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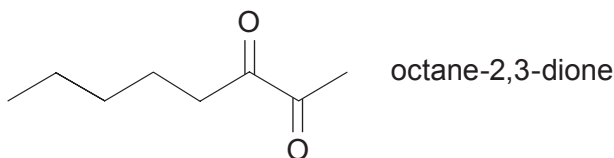
- (ii) Suggest how the rate of the reaction in part (i) could be studied. [1]

.....

.....



- (d) The mass spectrum of octane-2,3-dione (M_r 142) shows fragmentation peaks at m/z 99, 71 and 43.



Identify the species responsible for these peaks.

[2]

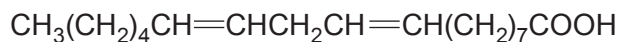
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- (e) Oct-1-en-3-ol is formed biochemically by the breakdown of linoleic acid.



linoleic acid

The carbon to carbon double bonds in linoleic acid are reduced by hydrogen and the resulting product can then be reduced by another suitable reducing agent to give a compound of formula $\text{C}_{18}\text{H}_{38}\text{O}$.

- (i) Write the **skeletal** formula of this compound.

[2]

- (ii) State the name of the reducing agent used in the second stage of the reduction.

[1]

.....

